Jeremy Demers

Full Stack Software Developer | Process Automation | AI Engineering

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PROFESSIONAL PROFILE

Jeremy Demers is a technically versatile full-stack software developer with a strong mix of enterprise application development, process automation, AI engineering, and real-world scientific/process experience. His background combines long-term enterprise software development with modern full-stack application work in pharmaceutical and laboratory environments. He builds practical systems: database-backed dashboards, APIs, internal data systems, automated reporting workflows, scientific integrations, computer-vision tools, and production-oriented applications that are useful, maintainable, and aligned with real operational needs.

He is especially strong at bridging software engineering with regulated scientific operations. His work spans React, Next.js, Django, Flask, Python, C#, ASP.NET Core, PostgreSQL, Oracle, Waters Empower 3, Docker, AWS deployment patterns, Terraform, GitHub CI/CD, reporting automation, data visualization, scheduled jobs, and AI/ML workflows. He is actively positioning for senior-level software engineering, scientific software, process automation, and AI-focused opportunities.

CAREER FOCUS

- Senior full-stack engineering roles that require practical architecture, maintainable systems, API design, data workflows, dashboards, and production-quality delivery.
- Scientific software, laboratory automation, SmartLab, LIMS-adjacent, and regulated-industry technology roles that benefit from both software and process background.
- AI-enabled application development, including computer vision, LLM-powered tools, RAG, agentic workflows, cloud-native AI systems, and AI-assisted full-stack development.

TECHNICAL CAPABILITIES

Full-Stack Development: React, Next.js, JavaScript, TypeScript, Bootstrap, Django, Flask, ASP.NET Core, REST APIs, Socket.IO, SQLAlchemy, data visualization, database-backed dashboards

Data Engineering and Reporting: PostgreSQL, Oracle, Waters Empower 3 APIs/database, SQLite, Pandas, OpenPyXL, Excel automation, recurring data collection, structured exports, instrument report parsing

AI and Computer Vision: YOLO-based detection, PyTorch, OpenCV, CNNs, DNNs, image models, local generative AI models, image captioning, diffusion models, GPU workflows

Cloud and DevOps: Docker, AWS application deployment, Terraform, GitHub Actions, CI/CD, Vercel, Azure, GCP, Bedrock, SageMaker, Lambda, API Gateway, S3, CloudFront, SQS, Aurora Serverless

Scientific and Regulated Process Knowledge: cGMP, FDA-regulated environments, chemical operations, process control, Lean 5S, biomanufacturing, biotech techniques, micropipetting, cell incubation, control-room simulation, mechanical/electrical fundamentals

PROFESSIONAL EXPERIENCE

Pfizer - Full Stack Web Developer and Process Automation | September 2020 - Present | Groton, CT

- Develop and maintain enterprise-style full-stack applications, internal data systems, process automation tools, reporting workflows, dashboards, APIs, and exports used in scientific and laboratory contexts.
- Build systems that connect to scientific and operational data sources, including PostgreSQL, Oracle, Waters Empower 3, instrument reports, usage summaries, scientist activity data, and recurring data-collection workflows.
- Create dashboards and visualization tools that turn laboratory and instrument data into accessible information for scientists and operational users.
- Automate recurring manual workflows through scheduled jobs, structured data transformations, Excel generation, file/report processing, and user-friendly web interfaces.
- Design practical application architecture with maintainability, traceability, production readiness, and enterprise-quality standards in mind.
- Apply a broad technical stack including Python, Django, Flask, React, Next.js, JavaScript, PostgreSQL, Oracle integration, Docker, APIs, data visualization, and deployment-oriented engineering practices.

Eurofins - Full Stack Web Developer and Process Automation | April 2019 - September 2020 | Groton, CT

- Built full-stack web applications and automation tools supporting laboratory, scientific, and process-oriented workflows.
- Worked with data-processing and reporting needs, turning instrument output and operational requirements into structured software solutions.
- Collaborated with process users and scientists to understand workflow pain points and convert them into practical, maintainable software features.
- Strengthened experience in regulated/scientific settings, process automation, data quality, and workflow-oriented software development.

MEDITECH - Lead Senior Software Developer | May 2000 - August 2015 | Westwood, MA

- Served in a senior/lead software development role for enterprise healthcare software in a large production environment.
- Developed, enhanced, troubleshot, and maintained long-lived enterprise software systems with a focus on reliability, maintainability, and workflow support.
- Collaborated across teams on requirements, implementation, debugging, testing, delivery, and support across the software development lifecycle.
- Built a strong foundation in enterprise engineering discipline, production support, structured development practices, user-facing workflow systems, and large-scale software maintenance.

Self-Employed - Independent Software Developer and Portfolio Builder | November 2025 - Present | Rhode Island / Remote

- Continue building public portfolio projects and technical assets across full-stack development, AI engineering, cloud architecture, developer tooling, GitHub repository structure, and production-quality application design.
- Use independent project work to demonstrate senior-engineering readiness through documented architecture, clean repositories, deployment practices, API design, testing, and real-world problem solving.
- Actively study and apply modern employer expectations for senior engineers, including AI-assisted development, cloud-native architecture, security posture, observability, and maintainable repo organization.

SELECTED PROJECTS

SmartLab Empower API

Technologies: C#, .NET 8, ASP.NET Core, Oracle.ManagedDataAccess.Core, Docker, Swagger

A lightweight, containerized ASP.NET Core API designed to retrieve analytical data from the Oracle Waters Empower 3 chromatography data system and expose it through a modern, structured interface for labs, LIMS integrations, SmartLab environments, and automated reporting workflows.

- Provides endpoints for sample set/result set pairs, time-range queries, sample name and year lookups, and result ID lookups.
- Modernizes access to Empower data by wrapping complex analytical data retrieval in a structured API layer.
- Supports local development through the .NET CLI and production/container deployment through Docker.
- Includes Swagger documentation for testing, validation, and easier adoption by technical users.
- Project purpose: streamline and modernize access to Empower 3 data for laboratory automation and SmartLab integration workflows.
- Repository: https://github.com/JeremyDemers/Empower-API

LLE Analysis

Technologies: Python, YOLO-based object detection, PyTorch, OpenCV, Flask, Socket.IO, Pandas, OpenPyXL, SQLite

An automated computer-vision system for analyzing liquid-liquid extraction samples as laboratory robotics move vials into camera view.

- Trained a custom YOLO-based convolutional neural network using more than 2,000 manually annotated images.
- Detects vial features and marks them with labeled bounding boxes for downstream analysis.
- Analyzes detections to determine liquid-layer positions and volumes, meniscus type, rag layers, residue, measurements, and sample viability.
- Generates detailed Excel reports that include structured results and annotated images.
- Automates delivery by emailing final results directly to scientists after runs complete.
- Demonstrates end-to-end AI application development: data labeling, model training, inference, image analysis, web/service integration, reporting, and workflow automation.
- Repository: https://github.com/JeremyDemers/LLE-Analysis

Disso Dissolution Platform

Technologies: Python, Django, PostgreSQL, Pandas, SQLAlchemy, JavaScript, Bootstrap

A full-stack web application that centralizes pharmaceutical dissolution study workflows and associated data management.

- Enables scientists to create, duplicate, edit, review, approve, and securely lock study records.
- Captures sample, method, media, instrumentation, analytical, and study-parameter details in a structured system rather than scattered manual records.
- Processes reports from Distek, ChemStation, Agilent, and ALMAC systems and transforms instrument output into structured data and Excel reports.
- Improves consistency, traceability, accessibility, and analysis readiness of dissolution study information.
- Includes a custom 3D-printer camera mount and lighting-system concept for the Cirius robot well-plate area.
- Demonstrates workflow-aware application design in a regulated laboratory context.

AI ENGINEERING AND MACHINE LEARNING DEVELOPMENT

- Completed a production-focused AI engineering course covering generative AI, agentic AI applications, LLM-powered SaaS apps, RAG, MCP-based agents, multi-agent workflows, cloud-native AI architecture, serverless infrastructure, Terraform, GitHub Actions, observability, security controls, guardrails, and cost management.
- Built hands-on exposure across Vercel, AWS, Azure, and GCP, with deeper AWS work involving Bedrock, SageMaker, Lambda, API Gateway, S3, CloudFront, SQS, and Aurora Serverless.
- Experimented with local generative AI models, image captioning, diffusion models, GPU-based workflows, CNNs, DNNs, and practical machine-learning application patterns.
- Approaches AI as a builder: emphasizing usable applications, real workflows, model integration, automation, data handling, deployment, observability, and guardrails rather than theory alone.

REGULATED SCIENTIFIC AND PROCESS BACKGROUND

In addition to software engineering, Jeremy has education and hands-on training in chemical operations, process control, Lean 5S practices, biomanufacturing, control-room simulation, biotech techniques, micropipetting, cell incubation, cGMP, FDA-regulated environments, and mechanical/electrical fundamentals. This background provides a practical bridge between software, laboratory operations, regulated processes, scientific users, and real-world manufacturing/process discipline.

EDUCATION

New England Institute of Technology: Associate Degree, Computer Science (1998 - 2000)

Community College of Rhode Island: Certificate, Chemical and Bio Process Technology (2019)

PORTFOLIO AND PUBLIC LINKS

LinkedIn: https://www.linkedin.com/in/jeremy-demers/
Portfolio: https://jeremysdemers.com/
GitHub: https://github.com/JeremyDemers
Portfolio Site Repo: https://github.com/JeremyDemers/portfolio-site
Empower API: https://github.com/JeremyDemers/Empower-API
LLE Analysis: https://github.com/JeremyDemers/LLE-Analysis
Digital Twin: https://github.com/JeremyDemers/digital-twin
Arcade: https://github.com/JeremyDemers/arcade

REPRESENTATIVE ROLE FIT

- Senior Full Stack Software Developer
- Scientific Software Developer
- Process Automation Engineer / Developer
- AI Application Engineer
- Computer Vision Application Developer
- Laboratory Automation / SmartLab Software Engineer
- Backend/API Developer with scientific data integration focus